IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

ROLE AND VALUE OF SMART GRID TECHNOLOGIES

Tuesday, 12 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions. All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : G. Strbac, F. Teng, F. Teng
Second Marker(s) : F. Teng, G. Strbac, G. Strbac

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Question 1

The three-bus bar electricity transmission system presented in the Figure 1.1, consists of two generation units, connected to busbars 1 and 3, which have a rating and a marginal cost (MC) as depicted in the figure. These two generators supply the loads located at buses 2 and 3.

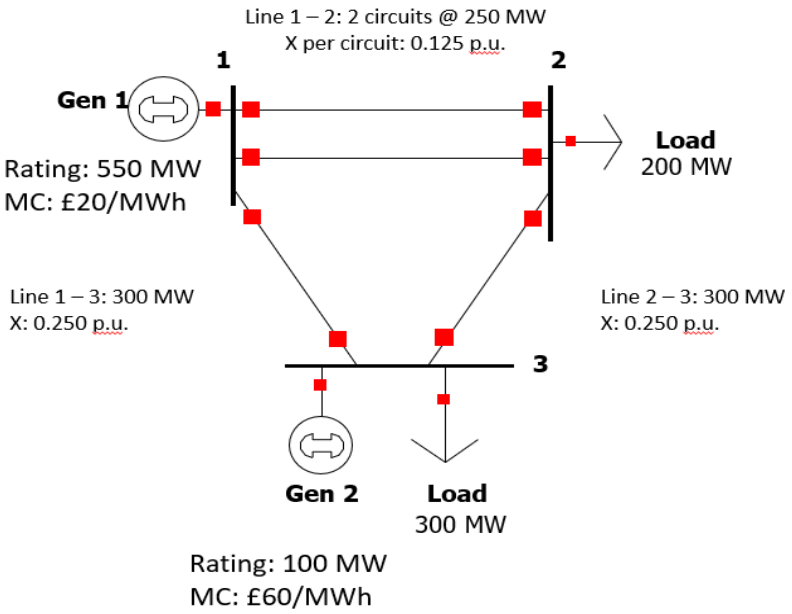


Figure 1.1 Three-bus bar electricity transmission system

Generation, demand and transmission circuit data are presented in corresponding tables below:

	Rating (MW)	Marginal Cost (£/MWh)
Gen 1	550	20
Gen 2	100	60

	Rating (MW)
Load (bus 2)	200
Load (bus 3)	300

	Rating per circuit (MW)	Reactance per circuit (pu)
Line 1-2	250	0.125
Line 1-3	300	0.250
Line 2-3	300	0.250

a) Determine the economic dispatch ignoring transmission line constraints. Calculate the corresponding cost and power flows.

[3]

b) Taking into account the capacity of transmission circuits, find the power flows when there is an outage at one of the lines as follows:

i) When line 1-3 is out of operation
ii) When line 2-3 is out of operation
iii) When one of the two circuits of line 1-2 is out of operation

[4]

c) For the case of question b), if you find that one of the lines is overloaded, re-dispatch generation to make sure that none of the lines would be overloaded. Calculate corresponding generation operation costs and explain the differences against a).

[5]

d) For the case of question b), determine the minimum Demand-Side Response (in busbar 2 or busbar 3) so that the power flow in the overloaded line are appropriately reduced in the case of circuit outages. Justify your answer. Calculate the corresponding generation operating costs.

[5]

e) Discuss the differences between preventive control solution c) against corrective control solution d).

[3]

Question 2

Part A

a) List the four primary sources of big data in power systems.

[2]

b) Explain how asset health monitoring was carried out in the past and how it can benefit from big data.

[3]

c) Explain how dynamic stability assessment is carried out by Transmission System Operators (TSOs) currently, and how data from Phasor Measurement Units (PMUs) can be used to improve this.

[3]

Part B

a) Explain how governments aim to decarbonise electricity systems. Which are the resulting challenges for these systems and how can demand flexibility contribute to addressing these challenges?

[3]

b) A simple power system, operating at a two-hour horizon, includes:
- a generator producing power s_t (MW) at hour t with a cost function $C_t(s_t) = 100s_t^2$ (£) and a maximum output limit $s^{max} = 10$MW
- inflexible demand appliances, consuming power $D_1 = 3.33$MW (at hour 1) and $D_2 = 1.67$MW (at hour 2)
- flexible demand consisting of 1000 identical appliances with continuously adjustable power d_t (kW), scheduling interval including hours 1 and 2, total energy required $E = 5$kWh and maximum power limit $d^{max} = 6$kW

i) Assuming that price-based scheduling of the flexible demand appliances is employed, with the prices at hour 1 and 2 being $\lambda_1 = 2£/kWh$ and $\lambda_2 = 1£/kWh$, what is the resulting power demand of each of the flexible demand appliances at hours 1 and 2? What is the total generation cost?

[3]

ii) Assuming that, apart from the prices λ_1 and λ_2, a flexibility penalty α (measured in $£/kWh^2$) is applied to the square of each flexible appliance's power demand, determine the resulting power demand of each of the flexible demand appliances at hours 1 and 2 as a function of α.

[3]

iii) Varying the flexibility penalty α in the range $0.1£/kWh^2$ to $0.5£/kWh^2$ with a step of $0.1£/kWh^2$, calculate its most suitable value from the system perspective, by employing a relevant matrix and explaining your calculations properly.

[3]

Question 3

a) Discuss how you might quantitatively characterise past and future performance for the distribution and transmission network. Provide examples of performance measures for the distribution and the transmission networks and discuss why they are different.

[3]

b) Consider a section of a distribution network serving 9500 customers. The outage events occurred in one year are reported in Table 3.1, except for an outage that affected 2000 customers scattered in a large area. In this last outage lasting for 1 hour the customers are not reconnected all at the same time. Assume that their reconnection on average followed a linear trend as a function of time. Compute CML and CI for this section of the network. Discuss the difference between CML and SAIDI.

Customers Affected	**Duration**
1200	30 minutes
300	5 hours
2500	2 minutes
900	50 minutes
20	36 hours
700	10 minutes

Table 3.1- Outage events

[3]

c) Consider the distribution network in Figure 3.1

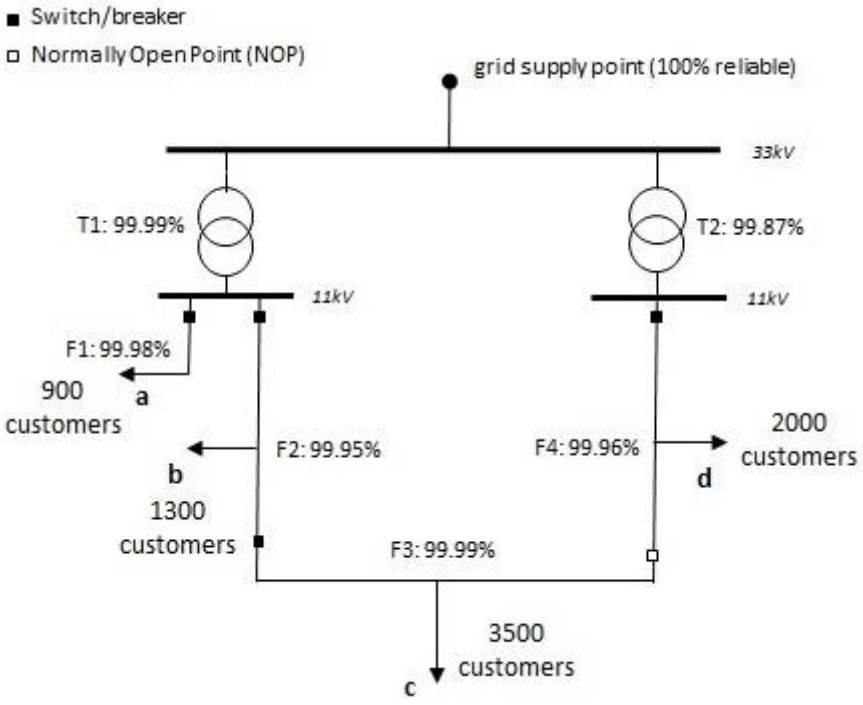


Figure 3.1- Network diagram

The number of customers at each load point (a, b, c, d) is reported in Figure 3.1. The availabilities of the transformers (T1, T2) and the feeders (F1, F2, F3, F4) are reported in Figure 3.1. Assume that all components failures are independent and they all last more than three minutes. The individual capacity of T1 and T2 are always sufficient

to serve all customers at load points a, b c and d. The network is radially operated but the normally open points (NOPs) can ensure some redundancy. They can be closed to resupply load points in case of faults in the network.

i) Construct reliability block diagrams for the power supply to each individual load point and calculate the unavailability of power in each case. Assume all interruptions are longer than 3 minutes. Determine the CML for this network.

[5]

ii) Assume that an interruption incentive scheme is in place and there is an adjustment on the revenues of the distribution company, based on the actual CML index and its target CMLTARGET=135 minutes. Such adjustment ΔR can be expressed as:

$$\Delta R = \text{k} * (CMLTARGET - CML) \qquad (£)$$

(constant k is assumed to be positive).

The company has the possibility of introducing switches S1 and S2 to isolate the transformers T1 and T2 from the bus B1 and B2 at 11k and place a new feeder connecting B1 to B2 with reliability $\rho = 99.99\%$ as illustrated in Figure 3.2. Assume that the reliability of the two new switches S1, S2 and the buses B1 and B2 can be approximated as 1. What would be the maximum cost of this upgrade that would justify the investment?

[6]

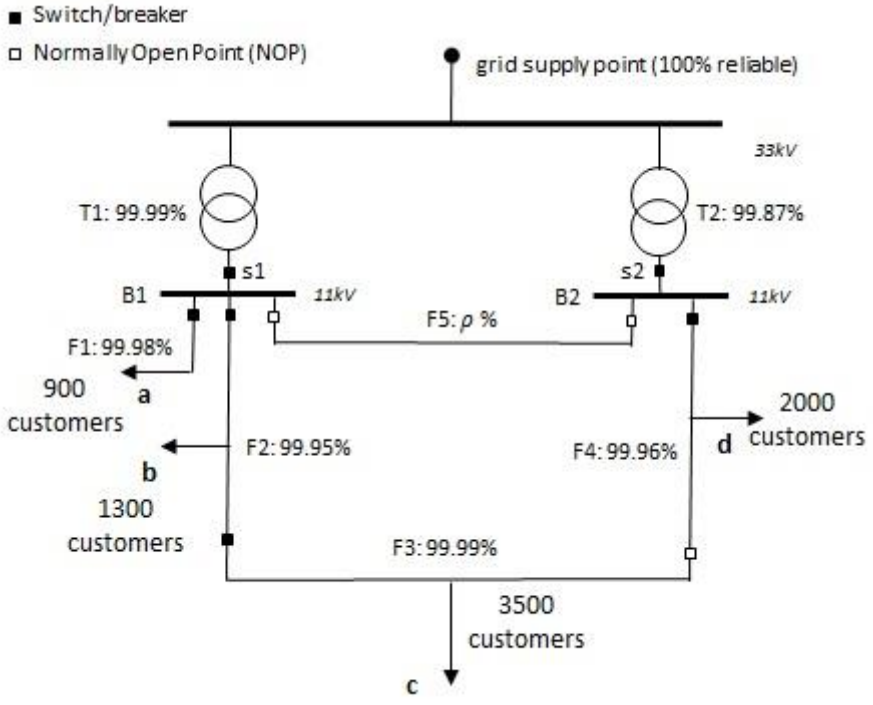


Figure 3.2: Modified Network

d) Consider a nonrepairable parallel structure of two independent components with lifetimes T_1 and T_2 independent and exponentially distributed with constant failure rates λ_1 and λ_2. Compute the mean time to failure of the parallel structure. Is the failure time T of the parallel structure exponentially distributed?
(Suggestion*: use the survival function* $S_T(t)$ *and note that* $[tS_T(t)]_0^\infty = 0$)

[3]

Question 4

a) Monte Carlo simulation is used to estimate the expected amount of interruptions.

i) The application of Monte Carlo Simulations relies on three fundamental theories. Please state these three theories and how they support the application.

[3]

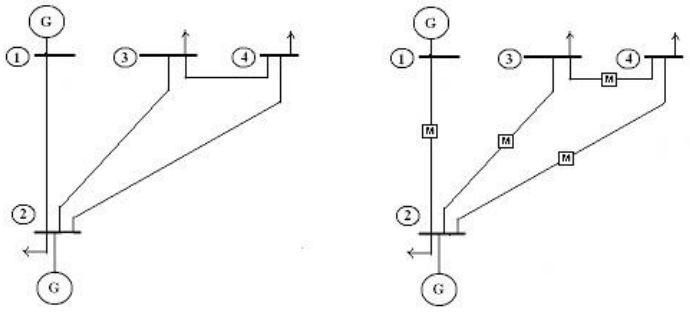
ii) After N=10,000 samples the best estimate of the interruption is 108,762 MW, with a standard error of 2581 MW. How many samples will be required (approximately) to reduce the coefficient of variation to 0.3%?

[2]

iii) What are the advantages and disadvantages of Monte Carlo simulation?

[2]

b) Consider the 4-bus system consisting of load and generators connected as below. The system is measured by meters placed on each branch. Assume PU reference is 1MW.



PER UNIT REACTANCES

X12	0.1
X23	0.2
X24	0.35
X34	0.5

PER UNIT POWER FLOWS

Z12	-70
Z23	37.6
Z24	32.4
Z34	7.6

Figure 4.1 4-bus system with branch meter placement and unit reactance

i) Construct the branch incidence matrix **H** for this system. To do this you will consider the incidence between each bus. Hint: Can be done either by direct inspection of the branches or by using a combination of the incidence **I** and branch diagonal **D** via **H=D∗I**.

[1]

ii) An attacker would like to convince the system operator that line Z23 is overloaded. The power flow limit for this line is 50MW. Assume bus 1 is the reference bus (θ_1=0), the estimated state for the original measurement are $\theta_2 = -7.00$, $\theta_3 = 0.524$, $\theta_4 = 4.333$. A naive attacking strategy is to change the measure of Z_{23} to 50 MW, which results in a new estimated states $\theta_2 = -7.00$, $\theta_3 = 2.86$, $\theta_4 = 5.8$. Please calculate the residuals for both the original measurements and the attacked measurements and comment on whether this attack strategy will be successful if the alarm on residual is set at 0.1.

[4]

iii) If the attack in ii) is not likely to be successful, how can the attacker design the attack to drive the overload without being detected? What is the minimum number of measurements the attacker needs to change in order to achieve this aim? Where appropriate, please use matrix formulations to justify your answer.

[5]

c) Explain the Willingness-to-Pay (WTP) and Willingness-to-Accept (WTA) approaches to evaluating customer interruption costs. Do you expect similar values when using WTA and WTP? Explain which approach you consider most suitable as a basis for interruption costs models.

[3]